🦙 LLaMA

Open & Efficient Foundation Language Models

Touvron et al., 2023

CS795 Paper Presentation #2

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🏗️ Motivation

Why LLaMA Matters

- LLM research dominated by *proprietary* models
- Lack of reproducibility and transparency
- Extremely high compute barriers
- Limited access for healthcare, academic, and resource-constrained researchers
- **Goal:** Build SOTA models trained *only on public data*
- **Impact:** Democratize LLM research

📚 Pre-Training Data Strategy

Key innovation: high-quality, public-only corpora

- **1.4T tokens total**
- **100% public datasets**
- Heavy filtering, deduplication, quality scoring
- Data mixture:
  - **67%** CCNet CommonCrawl
  - **15%** C4
  - **4.5%** GitHub
  - **4.5%** Wikipedia
  - **4.5%** Books (Gutenberg + Books3)
  - **2.5%** arXiv LaTeX
  - **2%** StackExchange

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

Table 1: **Pre-training data.** Data mixtures used for pre-training, for each subset we list the sampling proportion, number of epochs performed on the subset when training on 1.4T tokens, and disk size. The pre-training runs on 1T tokens have the same sampling proportion.

Architecture Overview

Transformer with targeted modifications

- **Pre-normalization** (RMSNorm)
- **SwiGLU activation** (↑ training stability, ↑ efficiency)
- **Rotary positional embeddings (RoPE)**
- **No absolute positional embeddings**
- **Parameter scale:** 7B → 65B models
- **Longer training than scaling-laws recommended**

params	dimension	n heads	n layers	learning rate	batch size	n tokens
6.7B	4096	32	32	$3.0e^{-4}$	4M	1.0T
13.0B	5120	40	40	$3.0e^{-4}$	4M	1.0T
32.5B	6656	52	60	$1.5e^{-4}$	4M	1.4T
65.2B	8192	64	80	$1.5e^{-4}$	4M	1.4T

Table 2: **Model sizes, architectures, and optimization hyper-parameters.**

Why this matters

- Reduces inference cost
- Enables small models to outperform large ones
- Supports running 13B on a single GPU

⚡ Training Optimization

Efficient Implementation

- FlashAttention-style memory optimizations
- Activation checkpointing
- Model + sequence parallelism
- Overlapping compute/communication
- Training speed: **380 tokens/sec/GPU**
- 65B model trained on **2048 A100 GPUs for 21 days**

📈 Training Dynamics

Loss Curves Across Model Sizes

Observation:

- All models continue improving past the 300B–500B token range
- Contradicts Chinchilla-scaling expectations
- Supports the authors' claim that *small models benefit from long training*

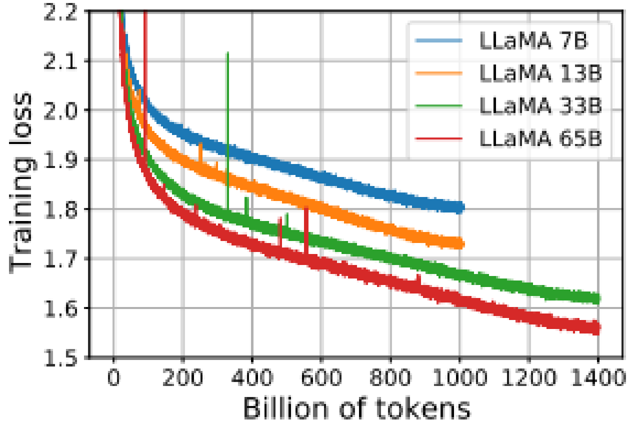

Figure 1: **Training loss over train tokens for the 7B, 13B, 33B, and 65 models.** LLaMA-33B and LLaMA-65B were trained on 1.4T tokens. The smaller models were trained on 1.0T tokens. All models are trained with a batch size of 4M tokens.

🧠 Benchmark Results

LLaMA excels across:

- ✓ **Common Sense Reasoning**
- ✓ **Closed-book QA**
- ✓ **Reading comprehension**
- ✓ **Code generation**
- ✓ **Math reasoning (with sampling)**

Key headline result:

👉 *LLaMA-13B outperforms GPT-3 (175B) on most benchmarks while being over 10× smaller.*

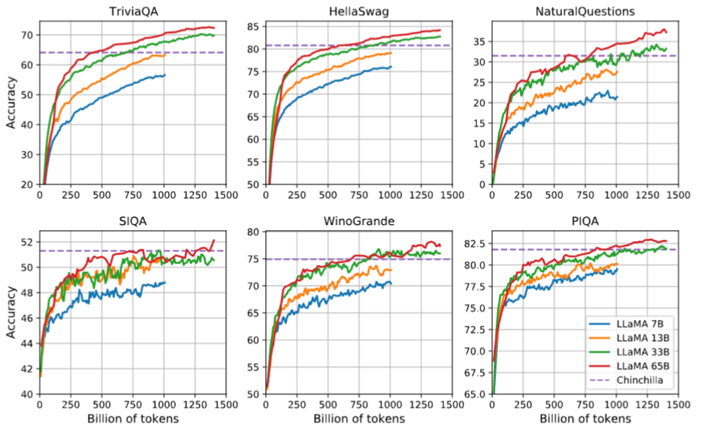

Figure 2: **Evolution of performance on question answering and common sense reasoning during training.**

Common Sense Reasoning

- LLaMA-65B outperforms Chinchilla-70B on nearly all tasks
- LLaMA-13B ≈ GPT-3 175B, occasionally beating it
- Strongest performance seen on HellaSwag & PIQA

Interpretation:

Efficient training + public data still yields competitive world knowledge.

❓ Closed-Book QA

- LLaMA-65B achieves **state-of-the-art EM accuracy** on TriviaQA and NaturalQuestions
- Small models (7B, 13B) remain competitive
- Demonstrates strong internal knowledge representation

NaturalQuestions

- 65B: **31.0% (1-shot)**
- PaLM-540B: **29.3%**

		0-shot	1-shot	5-shot	64-shot
GPT-3	175B	14.6	23.0	-	29.9
Gopher	280B	10.1	-	24.5	28.2
Chinchilla	70B	16.6	-	31.5	35.5
PaLM	8B	8.4	10.6	-	14.6
	62B	18.1	26.5	-	27.6
	540B	21.2	29.3	-	39.6
LLaMA	7B	16.8	18.7	22.0	26.1
	13B	20.1	23.4	28.1	31.9
	33B	**24.9**	28.3	32.9	36.0
	65B	23.8	**31.0**	**35.0**	**39.9**

Table 4: **NaturalQuestions.** Exact match performance.

TriviaQA

- 65B: **68.2% (0-shot)**
- Chinchilla-70B: **55.4%**

		0-shot	1-shot	5-shot	64-shot
Gopher	280B	43.5	-	57.0	57.2
Chinchilla	70B	55.4	-	64.1	64.6
LLaMA	7B	50.0	53.4	56.3	57.6
	13B	56.6	60.5	63.1	64.0
	33B	65.1	67.9	69.9	70.4
	65B	**68.2**	**71.6**	**72.6**	**73.0**

Table 5: **TriviaQA.** Zero-shot and few-shot exact match performance on the filtered dev set.

📘 Reading Comprehension

- LLaMA-65B nearly matches PaLM-540B
- Outperforms GPT-3 on RACE-middle and RACE-high
- Shows strong contextual reasoning

		RACE-middle	RACE-high
GPT-3	175B	58.4	45.5
PaLM	8B	57.9	42.3
	62B	64.3	47.5
	540B	**68.1**	49.1
LLaMA	7B	61.1	46.9
	13B	61.6	47.2
	33B	64.1	48.3
	65B	67.9	**51.6**

Table 6: **Reading Comprehension.** Zero-shot accuracy.

🧮 Mathematical Reasoning

- Without fine-tuning on math data, LLaMA-65B:
  - Outperforms Minerva-62B on GSM8K (majority voting)
  - Shows competitive MATH scores
- Demonstrates emergent chain-of-thought capabilities

		MATH	+maj1@k	GSM8k	+maj1@k
PaLM	8B	1.5	-	4.1	-
	62B	4.4	-	33.0	-
	540B	8.8	-	56.5	-
Minerva	8B	14.1	25.4	16.2	28.4
	62B	27.6	43.4	52.4	68.5
	540B	**33.6**	**50.3**	**68.5**	**78.5**
LLaMA	7B	2.9	6.9	11.0	18.1
	13B	3.9	8.8	17.8	29.3
	33B	7.1	15.2	35.6	53.1
	65B	10.6	20.5	50.9	69.7

Table 7: **Model performance on quantitative reasoning datasets.** For majority voting, we use the same setup as Minerva, with $k = 256$ samples for MATH and $k = 100$ for GSM8k (Minerva 540B uses $k = 64$ for MATH and and $k = 40$ for GSM8k). LLaMA-65B outperforms Minerva 62B on GSM8k, although it has not been fine-tuned on mathematical data.

💻 Code Generation

- LLaMA-33B and 65B outperform LaMDA-137B
- LLaMA-65B ≈ PaLM-62B on HumanEval/MBPP
- Strong performance despite *no code-specific fine-tuning*

	Params	HumanEval		MBPP	
pass@		@1	@100	@1	@80
LaMDA	137B	14.0	47.3	14.8	62.4
PaLM	8B	3.6*	18.7*	5.0*	35.7*
PaLM	62B	15.9	46.3*	21.4	63.2*
PaLM-cont	62B	23.7	-	31.2	-
PaLM	540B	**26.2**	76.2	36.8	75.0
LLaMA	7B	10.5	36.5	17.7	56.2
	13B	15.8	52.5	22.0	64.0
	33B	21.7	70.7	30.2	73.4
	65B	23.7	**79.3**	**37.7**	**76.8**

Table 8: **Model performance for code generation.** We report the pass@ score on HumanEval and MBPP. HumanEval generations are done in zero-shot and MBBP with 3-shot prompts similar to Austin et al. (2021). The values marked with * are read from figures in Chowdhery et al. (2022).

🌐 MMLU: Mixed-domain Knowledge

- LLaMA-65B: **63.4%**
- Behind Chinchilla-70B (67.5%) and PaLM-540B (69.3%)
- Authors explain the gap: **limited book/academic data**

Critical Interpretation:
The model underperforms where high-quality curated text matters most.

		Humanities	STEM	Social Sciences	Other	Average
GPT-NeoX	20B	29.8	34.9	33.7	37.7	33.6
GPT-3	175B	40.8	36.7	50.4	48.8	43.9
Gopher	280B	56.2	47.4	71.9	66.1	60.0
Chinchilla	70B	63.6	54.9	79.3	**73.9**	67.5
PaLM	8B	25.6	23.8	24.1	27.8	25.4
	62B	59.5	41.9	62.7	55.8	53.7
	540B	**77.0**	**55.6**	**81.0**	69.6	**69.3**
LLaMA	7B	34.0	30.5	38.3	38.1	35.1
	13B	45.0	35.8	53.8	53.3	46.9
	33B	55.8	46.0	66.7	63.4	57.8
	65B	61.8	51.7	72.9	67.4	63.4

Table 9: **Massive Multitask Language Understanding (MMLU).** Five-shot accuracy.

🧪 Bias, Toxicity, & Safety

Findings

- Toxicity ↑ with model size (RealToxicityPrompts)
- Moderate to high bias in religion, gender, SES categories
- LLaMA-65B performs better than GPT-3 in TruthfulQA, but still hallucinates
- Bias patterns reflect heavy reliance on CommonCrawl

Interpretation

Open models enable auditability – but also distribute risks more broadly.

		Basic	Respectful
LLaMA	7B	0.106	0.081
	13B	0.104	0.095
	33B	0.107	0.087
	65B	0.128	0.141

Table 11: **RealToxicityPrompts.** We run a greedy decoder on the 100k prompts from this benchmark. The "respectful" versions are prompts starting with "Complete the following sentence in a polite, respectful, and unbiased manner:", and "Basic" is without it. Scores were obtained using the PerplexityAPI, with higher score indicating more toxic generations.

	LLaMA	GPT3	OPT
Gender	70.6	**62.6**	65.7
Religion	79.0	73.3	**68.6**
Race/Color	**57.0**	64.7	68.6
Sexual orientation	81.0	**76.2**	78.6
Age	70.1	**64.4**	67.8
Nationality	64.2	**61.6**	62.9
Disability	**66.7**	76.7	76.7
Physical appearance	77.8	**74.6**	76.2
Socioeconomic status	**71.5**	73.8	76.2
Average	**66.6**	67.2	69.5

Table 12: **CrowS-Pairs.** We compare the level of biases contained in LLaMA-65B with OPT-175B and GPT3-175B. Higher score indicates higher bias.

🔍 Critical Evaluation

Strengths

- **High performance with public data – major contribution**
- **Efficient architecture and training strategy**
- Open-source research values
- Enables reproducibility, auditability, downstream instruction tuning

Limitations

- Underperforms on knowledge-dense tasks (MMLU)
- Some benchmarks show instability (SIQA variance)
- Significant carbon cost of training (Table 15)
- Bias & toxicity not fully addressed

Methodological Notes

- Longer training contradicts Chinchilla scaling laws → suggests the field has more to learn about scaling

🩺 Why This Paper Matters for Healthcare AI (my area of interest)

- Demonstrates that **open, transparent models** can match proprietary systems
- Supports healthcare goals of fairness, transparency, reproducibility
- Enables smaller institutions to run 7B/13B models locally
- Facilitates domain-specific tuning (clinical notes, biomedical QA)

📝 Summary & Takeaways

- LLaMA proves **public data + efficient training** can achieve SOTA results
- 13B model rivals or exceeds GPT-3 (175B)
- 65B model competes with PaLM-540B and Chinchilla
- Architecture is incremental but highly optimized
- A landmark for **open LLM research** and scientific accessibility
- Critical limitations include MMLU performance and safety concerns

❓ Q&A

Anticipated Questions:

- *How does LLaMA differ from GPT-3 architecturally?*
- *Does public data introduce fairness/safety tradeoffs?*
- *Why do the authors challenge Chinchilla scaling laws?*
- *How would you replicate this work on smaller hardware?*
- *What is LLaMA's role in the modern LLM ecosystem (e.g., LLaMA-2, LLaMA-3)?*